Super-Resolved 3-D Satellite Lidar Imaging of Earth via Generative Diffusion Models

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***Abstract*— Spaceborne lidars are essential for monitoring Earth's ecosystems, particularly in imaging forests, glaciers, and natural hazards. However, current satellite lidar systems, such as NASA's Global Ecosystem Dynamics Investigation (GEDI) and Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2), are limited in spatial resolution and photon density, constraining their ability to capture detailed surface topography and vegetation (STV) 3-D imagery. Airborne systems, such as NASA's G-LiHT, offer higher resolution but lack global coverage. To address these limitations, compressive satellite lidars (CS-Lidars) have been recently introduced, utilizing coded laser illumination and dynamic wavelength scanning for wide-field 3-D imaging. A novel framework, based on hyperheight data cubes (HHDCs), uses deep learning to transform sparse measurements into 3-D images, but its resolution remains constrained by the physical limitations of the instruments. This article proposes three approaches using generative diffusion models to achieve super-resolution lidar imaging, enhancing satellite data resolution. These methods involve learning conditional probabilities, guiding models via forward imaging, and leveraging high-resolution side information. The results show substantial improvements in the resolution of satellite lidar data, enabling fine-scale studies of forest structure and improving applications in forest management and environmental monitoring. The methodologies were tested in three regions of USA: Florida, Maryland, and California. The models were trained and tested on the first two, and their zero-shot capabilities were tested on the third, showing comparable results.**

***Index Terms*— Compressed sensing, diffusion models, inverse problem, lidar, super-resolution.**

NOMENCLATURE

AWSL	Adaptive wavelength scanning lidar.
CASALS	Concurrent artificially intelligent spectrometry and adaptive lidar system.
CHM	Canopy height model.
CHP	Canopy height profile.
CS-Lidar	Compressive satellite lidar.
DDPM	Denoising diffusion probabilistic models.
DPS	Diffusion posterior sampling.
DTM	Digital terrain model.
FWHM	Full-width at half-maximum.
GEDI	Global Ecosystem Dynamics Investigation.
GT	Ground truth.
G-LiHT	Goddard's lidar, hyperspectral, and thermal system.
HHDC	Hyperheight data cube.
ICESat-2	Ice, Cloud, and Land Elevation Satellite-2.
KL	Kullback–Leibler (divergence).
Lidar	Light detection and ranging.
NEON	National ecological observatory network.
NSF	National Science Foundation.
OSSE	Observing system simulation experiments.
PSNR	Peak signal-to-noise ratio.
SERC	Smithsonian environmental research center.
SSIM	Structural similarity index measure.
STV	Surface topography and vegetation.

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Data is available on-line at https://github.com/Anfera/LidarDiffusionSuperRes.

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I. INTRODUCTION

SPACEBORNE lidars play a pivotal role in monitoring the Earth's intricate ecological processes. By imaging forests, they reveal the topological structures that significantly impact nutrient, water, and carbon cycles. They are crucial for glacier characterization and identifying natural hazards like landslides, volcanoes, and earthquakes [1], [2], [3]. Current spaceborne lidars, such as NASA's GEDI [4] and ICESat-2 [5], face challenges in spatial resolution. GEDI, for instance, captures laser reflections within large footprints (approximately 25 m in diameter), limiting the detail of the imaging. In addition, ICESat-2 struggles with acquiring a sufficient number of photons due to the vast distance to its target, which further restricts the quality of the data collected. In contrast, airborne lidars, such as NASA's G-LiHT [6], mounted on fixed-wing aircraft at lower altitudes, provide photon-dense, high-resolution point clouds. The disparity between ICESat-2's sparse line profiles and G-LiHT's rich 3-D imagery underscores the need for satellite lidars with comparable resolution and coverage. Such advancements would revolutionize Earth science and other applications.

To overcome these challenges, a novel approach is being developed by a new generation of lidars, such as those in NASA's CASALS program, built on the AWSL technology [7], [8]. Instead of the traditional 1-D line scanning method which satellite lidars take, CASALS introduces the concept of coded illumination in 2-D, coined CS-Lidars [9]. Unlike the standard

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practice of measuring 1-D line profiles along a satellite's swath path, CS-Lidars use sparse coded laser illumination across a 2-D area. This is made possible by tunable lasers and a dispersive grating. Through dynamic shifts in laser wavelengths, a vast 2-D coding matrix of laser footprints and spacings is created during the satellite's flight path. Wide field-of-view 3-D landscapes of Earth can then be reconstructed through computational techniques using the acquired samples [9], [10].

Given the CS-Lidar sensing framework, a novel methodology for the reconstruction of lidar imagery was recently introduced in [9]. It is based on the data representation called the HHDC, as described in Section II. By harnessing the capabilities of deep learning techniques, this methodology learns the intricate interplay between features derived from the sparse satellite measurements and transforms them into 3-D imagery that is comparable with imagery obtained by low-altitude systems [10], [11], [12]. This approach significantly enhances the performance of satellite-based technologies improving photon resolution and allowing sparse footprint sampling for optimal energy utilization. However, despite these advancements offered by HHDC representations for CS-Lidars, their ultimate resolution remains constrained by the physical characteristics of the instrument. Specifically, the illumination pattern, footprint size, and vertical resolution dictated by the instrument's resolution play a pivotal role. Therefore, expanding the capabilities of this framework through super-resolution algorithms is crucial.

Super-resolution algorithms could overcome the limitations imposed by the physical constraints of lidar instruments, unlocking new possibilities for high-resolution imaging. This advancement offers great potential for various applications that previously were limited to lidars mounted on drones or airplanes at low altitudes but limited in scalability. Super-resolution methods developed in this article could empower researchers to study forest structure at a finer scale, enabling them to discern individual tree crowns, canopy gaps, and other subtle features all from satellite measures [13]. This enhanced understanding can lead to more accurate estimations of forest biomass, carbon storage, and biodiversity, ultimately informing better forest management practices [14]. By enabling the detection of subtle changes in forest structure over time, it could improve researchers' ability to monitor deforestation, forest degradation, forest regrowth, and habitat mapping [15], [16], [17].

Our approach is based on recent generative diffusion models that have shown unparalleled performance in a vast number of applications [18], [19]. Inspired by the natural phenomenon of diffusion, these models work by first gradually adding noise to a high-resolution image, progressively transforming it into pure noise. Then, they learn to "reverse" this process, iteratively removing the noise and reconstructing the original high-resolution image. This iterative approach allows diffusion models to learn complex data distributions that can then be used to sample images and generate exceptionally high-quality outputs. These models have demonstrated remarkable performance in diverse domains, surpassing state-of-the-art technologies such as generative adversarial networks. For example, these models have been widely used in medical imaging, astronomical observation, and remote sensing, where sharper details and clearer images can lead to earlier diagnoses, groundbreaking discoveries, and more informed environmental management [20], [21], [22]. However, diffusion models can be difficult to guide, in that they might not be able to generate data that satisfies certain conditions or constraints imposed by the problem. This is particularly true in super-resolution problems since multiple high-resolution images may be consistent with a single low-resolution image [23]. Therefore, there is the urge to develop methods that can effectively steer the diffusion models in a meaningful way in order to enhance the resolution of satellite lidar data.

In this work, we propose three distinct approaches for achieving super-resolved lidar imaging. First, we learn the conditional probability distribution that relates low-resolution HHDCs to their high-resolution counterparts, enabling a supervised system to train the diffusion model. This allows the model to utilize all features of the actual lidar shots during the reverse denoising process. Second, we guide the diffusion model with a forward imaging model, which aligns with NASA's OSSEs. OSSEs use computer models to estimate the impact of proposed satellite systems before deployment. By simulating the satellite's sensing process, we assess the influence of satellite observations on environmental monitoring, helping to design and enhance new lidar systems. Similar to compressed sensing, this method provides a closed-form mathematical expression connecting low- and high-resolution measurements. This connection, along with the gradients of a cost function, guides the reverse diffusion process. Finally, we utilize side information with a native resolution that matches the high-resolution HHDCs. This side information serves as an example for the diffusion model, aiding in generating accurate high-resolution images. While our methodology leverages advanced AI techniques like diffusion models, it is grounded in the physics of lidar sensing. Rather than generating arbitrary or hallucinated outputs, our algorithms integrate meaningful physical information, such as the inherent sparsity of the lidar signals and the characteristics of forests [24], [25]. This ensures that the reconstructions remain consistent with real-world phenomena. By incorporating these constraints, our approach yields reliable and interpretable results, in contrast to purely data-driven methods that may overlook the physical realities of the sensed environment.

The main contributions of this article are as follows.

1) We introduce a new forward imaging model for satellite lidar systems, emulating the instrument's sensing process by analyzing Gaussian beams, footprint aggregation, and photon loss dynamics.
2) We formulate a Bayesian compressed sensing problem using a maximum likelihood approach to guide diffusion models for super-resolution in satellite lidar systems.
3) We provide a detailed methodology to significantly enhance the resolution of satellite lidar tensors in the context of HHDCs, leveraging diffusion models.
4) We examine how various modifications to the guidance systems of diffusion models improve super-resolution

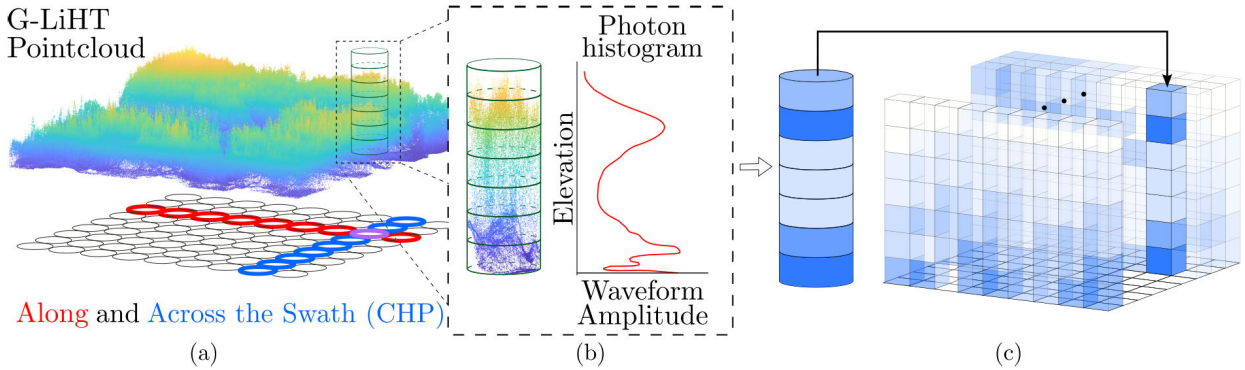


Fig. 1. HHDC. (a) High-resolution point cloud featuring a tube-shaped illuminated footprint is shown within a grid of multiple footprints, arranged along and across the satellite's trajectory. This arrangement simulates the illumination pattern of a satellite lidar system. (b) Histogram of return heights within the footprint represents the equivalent output waveform of a linear-mode lidar with single-photon sensitivity, similar to what is anticipated for the future CASALS mission. (c) Concatenated waveforms form the 3-D HHDC, where each voxel represents the photon count, creating a high-resolution representation of the scene, and here, darker shades of blue correspond to higher photon counts.

and demonstrate the combined benefits of these adjustments.

5) We show how these novel methods can be applied to HHDCs of forested regions in two distinct regions in U.S. territory.

The rest of this article is organized as follows. Section II presents an overview of the HHDC representation. Section III presents the physical foundations of the sensing phenomena of compressive satellite lidars. This allows us to pose a forward imaging model as well as a compressed sensing problem in the context of HHDCs. Section IV presents the ideas behind diffusion models and how they can be modified to implement super-resolution algorithms. Section V shows how diffusion models can be guided using various mechanisms. Section VI presents a detailed comparison of the different super-resolution systems on two forested areas in USA. Finally, Section VII presents the conclusion and proposes new future directions for this research.

II. Hyperheight Data Cubes

An HHDC is a novel method for representing lidar data, specifically designed to capture the 3-D structural information of landscapes. This approach is particularly useful for ecological studies, such as forest canopy profiling and digital terrain modeling. Unlike traditional 2-D lidar profiles, which provide only a cross-sectional view along a satellite's path, HHDCs capture a full volumetric data cube, encoding the length, width, and height of the observed area [9]. This 3-D structure allows researchers to analyze detailed spatial information, making it highly valuable for studying vegetation and terrain, and applying advanced deep learning techniques.

To construct an HHDC, consider the high-resolution point cloud depicted in Fig. 1(a), which highlights a single cylindrical footprint within a grid of footprints. The cylindrical footprint represents an illuminated area whose diameter matches the satellite lidar's laser shot size. Within each cylindrical footprint, as highlighted in Fig. 1(a), the point cloud data are processed to generate a histogram of photon return heights. This process involves dividing the vertical space within the footprint into discrete elevation bins. Each bin represents a specific height range above the ground, and the number of photon returns within that range is counted to determine the bin's value. This histogram, shown in Fig. 1(b), effectively captures the vertical distribution of lidar returns and emulates the waveform output of a linear mode with single-photon sensitivity lidar system, such as those anticipated for future missions like CASALS. The process is then extended to multiple footprints that are spaced along and across the swath, forming a grid. The histograms from each footprint are aligned based on their spatial positions and concatenated into a 3-D tensor structure, as illustrated in Fig. 1(c). This tensor, the HHDC, organizes the data such that the first two dimensions correspond to the spatial layout of the footprints (length and width of the swath), while the third dimension encodes the vertical distribution of photon returns. This compact and structured representation encapsulates the full 3-D structural information of the scene, enabling detailed ecological and topographic analyses.

The HHDC is represented as a three-order tensor $\mathcal{X} \in \mathbb{R}^{n\times m\times c}$, where n and m correspond to the number of footprints along and across the swath, respectively, and c is the number of vertical bins in each histogram. Each voxel in this cube encodes the photon count at a specific spatial location and height, with darker shades of blue in the visualization indicating higher photon densities. This flexible representation not only enables the extraction of traditional ecological metrics such as CHMs and DTMs but also facilitates multidimensional analysis, such as vertical slices along and across the swath, to provide a more comprehensive view of the landscape.

A notable advantage of HHDCs is their inherent sparsity in wavelet representations. Typically, less than 5% of coefficients are significant in natural forestry scenes, making them well suited for compressive sensing. This property allows efficient capture of essential 3-D information with fewer measurements, which is especially critical for satellite-based systems operating under strict energy and bandwidth constraints. For example, the CASALS system uses overlapping footprints to maximize photon capture and employs HHDCs to consolidate this information into a format suitable for advanced ecological and topographic studies. To address the challenges posed by sparse sampling, machine learning techniques, such as 3-D convolutional autoencoders (CAEs), are employed

to reconstruct high-resolution HHDCs from compressively sampled data. These neural networks not only fill in missing data but also enhance the resolution of photon counts, enabling the generation of HHDCs with features comparable to those captured by airborne systems. This capability bridges the gap between satellite and airborne lidar, allowing high-resolution analysis on a global scale. The versatility of HHDCs extends beyond ecological applications. They can support hydrological modeling, cryospheric research, and other studies requiring detailed 3-D spatial information. Their capacity to represent and analyze large-scale terrains in a compact, sparse, and highly informative format positions HHDCs as a cornerstone for next-generation lidar remote sensing.

Furthermore, the HHDC framework and the reconstructions presented here are tailored to address the finer spatial scales and continuous coverage anticipated in future satellite lidar missions, such as NASA's CASALS [7]. While it is theoretically possible to adapt these methodologies to current satellite lidar systems like GEDI, the larger footprints and sparser sampling patterns of these instruments present significant challenges for the direct application of HHDC techniques. Instead, our approach is forward-focused, aligning with the evolving capabilities of next-generation satellite lidar systems designed to deliver denser, higher resolution data. This vision underscores the relevance of the proposed super-resolution techniques, which are poised to unlock unprecedented insights into forest structures and enhance environmental monitoring as these advanced technologies become operational.

III. Inverse Problem and Physical Modeling of Satellite Lidar Systems

Super-resolution in CS-Lidars can be posed as an inverse problem. In this case, we have a set of partial measurements $\mathbf{y}$ that came from $\mathbf{x}$, where the mapping $\mathbf{x} \mapsto \mathbf{y}$ is many-to-one, and therefore, it is a highly ill-posed problem. Let $\mathcal{X}_l \in \mathbb{R}^{c\times n\times m}$ be the tensor of low-resolution measurements at the satellite, where n and m are the along and across the swath dimensions, respectively, and c is the height dimension. Let $\mathcal{X}_h \in \mathbb{R}^{c_2\times n_2\times m_2}$ be the high-resolution tensor we wish to recover, where $n_2 > n$, $m_2 > m$, and $c_2 > c$ are the corresponding high-resolution dimensions. The forward imaging model for CS-Lidars can be described as

$$\mathcal{X}_l \sim \mathcal{A}(\mathcal{X}_h; r, \mathcal{G}) \tag{1}$$

where $\mathcal{A}(\cdot\,; r, \mathcal{G}) : \mathbb{R}^{c_2\times n_2\times m_2} \mapsto \mathbb{R}^{c\times n\times m}$ represents the probabilistic distribution governing the sensing process of the satellite lidar system (see Fig. 2). Given that $\mathcal{A}(\cdot)$ is a probability distribution, we state that the tensor $\mathcal{X}_l$ is distributed as a function of $\mathcal{X}_h$. This model is defined by two key parameters: r, which denotes the number of photons returning to the satellite per footprint, and $\mathcal{G}$, a kernel that defines the footprint dimensions on the ground as captured by the satellite. The forward imaging model plays a key role in the diffusion generative models we develop since the underlying physics of $\mathcal{A}(\cdot)$ can be used to constrain the solutions of the inverse problem, hence avoiding the need of sparsity conditions and successfully guiding the reverse diffusion process. Three key phenomena arise in the sensing mechanism of a CS-Lidar.

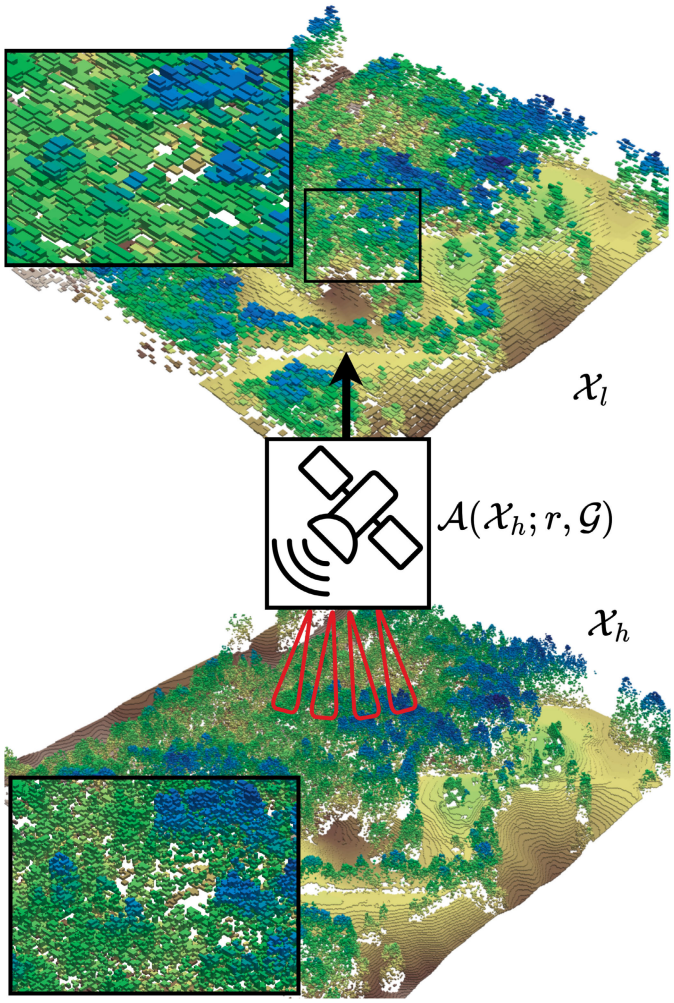


Fig. 2. Super-resolution of satellite lidars uses low-resolution measurements $\mathcal{X}_l$ in order to reconstruct a high-resolution tensor $\mathcal{X}_h$. This problem can be seen as an inverse problem in which we know the mapping $\mathcal{X}_l \sim \mathcal{A}(\mathcal{X}_h; r, \mathcal{G})$ that completely describes the sensing mechanism of the instruments in orbit.

1) *Gaussian Beam:* The laser shots exhibit a Gaussian beam pattern, determining the dimensions of the footprints on the ground (Fig. 3).
2) *Footprint Aggregation:* The footprint superposition patterns (Fig. 4).
3) *Photon Loss With Distance:* As the lidar instrument moves farther from Earth, there is a reduction in the number of detected photons (Fig. 5).

A. Gaussian Beams

Gaussian beams describe the primary source of laser illumination due to their well-defined intensity profiles and propagation characteristics. The bell-shaped intensity distribution of Gaussian beams ensures a concentrated beam with maximum intensity at the center, facilitating precise targeting of distant objects and efficient use of optical power. Their intensity profile as a function of distance z is described by [26]

$$I(x, z) = \frac{2P}{\pi w(z)^2} \exp\left(-2\frac{x^2}{w(z)^2}\right) \tag{2}$$

where P is the optical power of the laser, x is the radial distance from the beam axis, and $w(z)$ is known as the

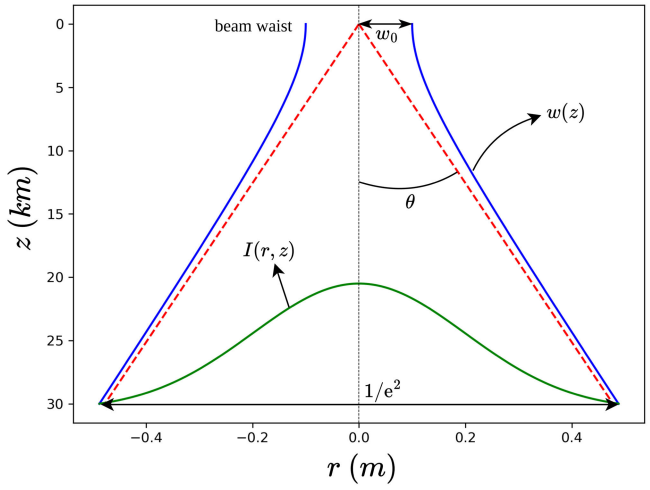


Fig. 3. $w(z)$ describes how the radius of the Gaussian beam changes as the distance increases. This parameter determines the footprint size, as it describes the $1/e^2$ distance for the intensity curve $I(r, z)$ and, therefore, the FWHM of the shot. As z increases, the beam radius can be approximated by the beam divergence θ.

beam radius (Fig. 3). The beam radius represents the size of the Gaussian beam at any given distance z from the beam waist. It is typically defined as the distance from the beam axis at which the intensity drops to $1/e^2$ (about 13.5%) of its maximum value. Diffraction causes the beam radius to change with the propagation direction, instead of staying constant. The beam radius is smallest at the beam waist and increases as the beam propagates away from it. The formula that describes the relationship between the beam radius and the distance z is given by $w(z) = w_0(1 + (z/z_R)^2)^{1/2}$, where w_0 is the radius at the beam waist and $z_R = \pi w_0^2/\lambda$ is the Rayleigh length (with λ being the wavelength of the laser). Fig. 3 depicts the shape of the beam radius as a function of the distance z, alongside the Gaussian intensity curve at a fixed distance.

Given w_0, we can determine how the radius of the beam changes along z, and therefore, we can calculate the physical dimensions of the lidar footprint. As we will see in Section III-D, the parameters that are relevant for the forward imaging model are the FWHM and the $1/e^2$ distance of the footprint; these parameters are described as: $1/e^2 = 2w(z)$ and $\text{FWHM} = w(z)\sqrt{2\ln 2}$. In many cases, the distance between the laser source and the object to be measured is so large that this phenomenon can be simplified to a single parameter θ called laser divergence (Fig. 3) so that the footprint radius can be determined by a simple trigonometry relationship using the satellite's altitude and the laser divergence.

B. Footprint Agregation

The 2-D lidar illumination patterns can be modeled as an aggregation of footprints, where multiple high-resolution lidar shots (taken at low altitude) overlapping over a given area are aggregated into a single low-resolution shot (taken at high altitude), as depicted in Fig. 4. The footprint aggregation phenomena model the loss of spatial resolution as we increase the distance from Earth by sampling bigger patches on

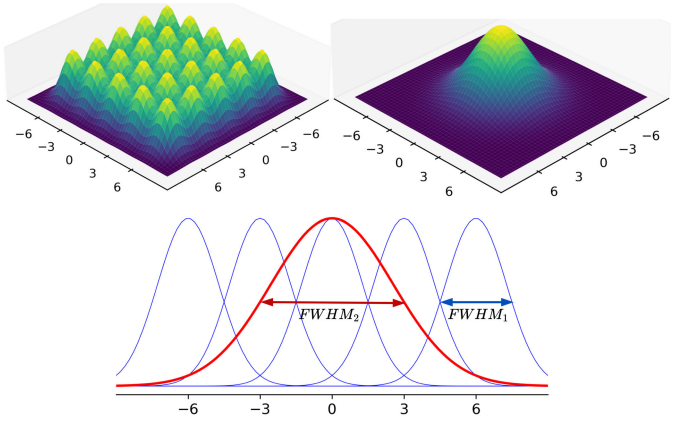


Fig. 4. Footprint aggregation is described by the superposition of multiple Gaussian beams that coincide within the $1/e^2$ of the low-resolution shot. In this example, 25 high-resolution footprints are aggregated into a single low-resolution footprint.

ground. This aggregation process is similar to downsampling operations where the goal is to reduce resolution by merging neighboring pixels. However, downsampling 3-D imagery of CS-Lidars is more complex due to the Gaussian nature of a lidar footprint. In the HHDC framework, we already know the waveforms generated from the high-resolution shots (referred to as $\mathcal{X}_h$). Therefore, it is essential to describe how to aggregate them while considering the FWHM of the low-resolution satellite shot.

Given that the energy distribution on ground has a Gaussian shape, the aggregation mechanism can be modeled by the convolution of the high-resolution HHDC $\mathcal{X}_h$ with a 3-D Gaussian kernel described by: 1) the FWHM of the low-resolution shot; 2) the number of high-resolution shots that coincide with the low-resolution shot (Fig. 4); and 3) the vertical resolution of the low resolution system (Fig. 5). In a 2-D space, a 2-D Gaussian filter can be expressed as

$$G(i, j) = \frac{1}{2\pi\sigma_l^2} \exp\left(-\frac{(i - k/2)^2 + (j - k/2)^2}{2\sigma_l^2}\right) \tag{3}$$

where i and j are the integer indices of the element in the kernel, ranging from 0 to $k - 1$; k is the size of the kernel described by the amount of high-resolution footprints that coincide with the low-resolution footprint; and $\sigma_l = \text{FWHM}_l/(2(2\ln 2)^{1/2})$ is the standard deviation controlling the spread of the bell-shaped curve. Finally, to consider the loss in vertical resolution, it is necessary to repeat the 2-D kernel $\eta = c_2/c$ times on the vertical axis, which effectively averages the vertical bins and arrives to the 3-D kernel $\mathcal{G} \in \mathbb{R}^{\eta\times k\times k}$.

This implies that to obtain the aggregated version $\mathcal{X}_a$ and thus model the footprint aggregation phenomena, we must follow the operation:

$$\mathcal{X}_a = \mathcal{G} * \mathcal{X}_h \tag{4}$$

where $*$ represents a strided convolution with stride $[\eta, n_2/n, m_2/m]$.

C. Photon Loss With Distance

The loss of photons is another factor that affects the resolution of an HHDC, and it cannot be accurately modeled

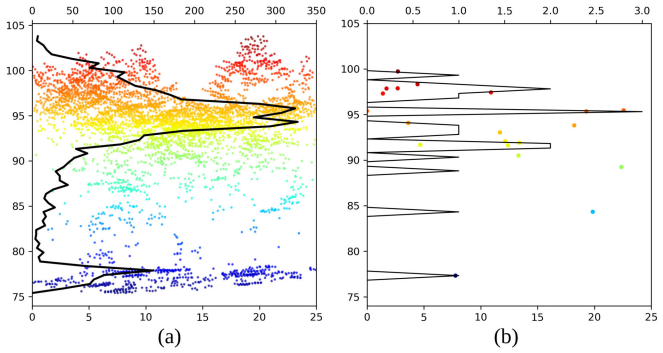


Fig. 5. (a) Thousands of photons in the high-resolution footprint and the corresponding histogram waveform. (b) Tens of photons in the low-resolution footprint and the corresponding histogram waveform.

through a 3-D convolution with a Gaussian kernel. As the lidar's altitude increases, fewer photons return to the detector. This reduction in photon count leads to a degradation in the definition of the satellite waveform, as depicted in Fig. 5. The connection between a high-resolution waveform and its low-resolution counterpart is closely tied to the number of photons detected by the lidar, which is directly proportional to the energy returning to the satellite. The number of detected photons can be expressed as $r = E_{\text{det}}\lambda/(ch)$, where E_{det} represents the energy sensed by the detector, λ is the laser's wavelength, c is the speed of light, and h is Planck's constant. Two key physical phenomena further characterize this process.

1) *The Photon Attenuation:* $\tau = \exp^{-\int_0^z \gamma(z,\lambda)dz}$, which governs the rate of photon extinction as the bundle of transmitted photons in a laser pulse travels from the Earth's surface to the sensor. The attenuation is caused by molecular absorption and particle scattering modeled in $\gamma(z, \lambda)$.
2) *The Isotropic Surface Reflection:* ρ that determines how much energy is lost due to the absorption and dispersion of photons by the Earth's surface. The photon reflection density quickly drops with a factor of z^2, where z is the distance from the target to the sensor [10].

These phenomena are encapsulated in the lidar equation that relates the transmitted pulse energy (E_{shot}) with the detected photon distribution energy (E_{det})

$$E_{\text{det}} = E_{\text{shot}} \frac{QA\rho\tau^2}{2\pi z^2} \tag{5}$$

where Q represents the detector's quantum efficiency and A represents the area of the lidar telescope, which determines the diffraction limit and the resolution of the optical system. The altitude of the lidar holds significant sway over the energy detected in the returning laser pulse. For instance, for the CS-Lidar in [7] that is going to be deployed at around 500 km, it is expected to have around 20 photons back per shot.

The relationship between high- and low-resolution lidar waveforms can be modeled by considering the random selection of returning photons due to atmospheric attenuation at higher altitudes. Specifically, at lower altitudes, each laser shot with total energy E_{shot} produces a high-resolution waveform that indicates the number of photons reflected and

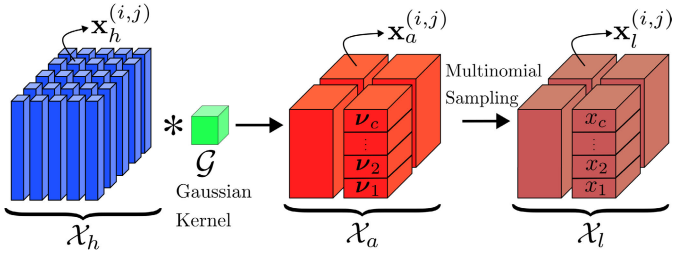


Fig. 6. High-resolution footprints, denoted as $\mathbf{x}_h$, within a high-resolution tensor $\mathcal{X}_h$, are convolved with a Gaussian kernel $\mathcal{G}$ to produce aggregated footprints $\mathbf{x}_a$, encapsulated in the aggregated tensor $\mathcal{X}_a$. By applying multinomial sampling to each aggregated footprint, based on the photon count r, we derive low-resolution footprints $\mathbf{x}_l$, represented within the low-resolution tensor $\mathcal{X}_l$.

received from various altitude bins. This waveform can be used to establish a probability distribution, where the normalized amplitude ν_k represents the likelihood of receiving a photon from altitude bin k. To simulate the effects of increased altitude and atmospheric attenuation, we assume that photon selection can be approximated as a probabilistic process and model it using multinomial sampling. Under this assumption, a total of r photons that return to the satellite at higher altitudes are sampled from this distribution, effectively reducing the high-resolution waveform to a low-resolution waveform. The resulting low-resolution waveform is assumed to follow a multinomial distribution with probability mass function

$$f(x_1, \ldots, x_c; r, \nu_1, \ldots, \nu_c) = \frac{r!}{x_1! \cdots x_c!} \nu_1^{x_1} \cdots \nu_c^{x_c} \tag{6}$$

where r is the total number of photons returning to the satellite, ν_k is the normalized amplitude (probability) for altitude bin k, and x_c is the number of photons counted in each bin, with $\sum_{i=1}^{c} x_i = r$ and $\sum_{i=1}^{c} \nu_i = 1$. By modeling the photon loss from a single footprint using this multinomial distribution, we can extend the approach to multiple laser shots, such as those in HHDCs, by employing independent multinomial distributions for each footprint. The results can then be consolidated into a single tensor $\mathcal{X}_l$, which encapsulates the aggregated photon return data from all footprints. While this approach simplifies the representation of atmospheric attenuation, it provides a statistically consistent and tractable framework for modeling photon selection in satellite lidar systems.

D. Bayesian Compressed Sensing

The combination of the three previously described phenomena gives the comprehensive definition of the forward imaging model for CS-Lidars (Fig. 6)

$$\mathcal{X}_l = \underbrace{\begin{pmatrix} \mathbf{x}_l^{(1,1)} & \cdots & \mathbf{x}_l^{(1,m)} \\ \vdots & \ddots & \vdots \\ \mathbf{x}_l^{(n,1)} & \cdots & \mathbf{x}_l^{(n,m)} \end{pmatrix}}_{\in \mathbb{R}^{c \times n \times m}} \sim \mathcal{A}(\mathcal{X}_h; r, \mathcal{G})$$

where $\mathcal{A}(\cdot\,; r, \mathcal{G}) : \mathbb{R}^{c_2 \times n_2 \times m_2} \mapsto \mathbb{R}^{c \times n \times m}$ is a probability distribution that combines the Gaussian nature of the beams, the footprint aggregation phenomena, and the photon loss.

Here, $\mathbf{x}_a^{(i,j)} \in \mathbb{R}^c$ is a vertical tube of the 3-D tensor of aggragated footprints, while $\mathbf{x}_l^{(i,j)}$ is a vertical tube of the low-resolution 3-D tensor, following a multinomial distribution $\mathbf{x}_l^{(i,j)} \sim \text{Multinomial}(r; \mathbf{x}_a^{(i,j)})$. Finally, $\mathcal{X}_a = \mathcal{G} * \mathcal{X}_h$ is the convolution that models the footprint aggregation phenomena, and r is the number of photons. This forward model assumes an ideal case without considering noise, both in the low- and high-resolution tensors. While the inclusion of noise would more closely reflect real-world scenarios, this complexity is reserved for future work, where its influence on reconstruction performance and mask optimization will be systematically addressed.

In the context of traditional compressed sensing, our objective is to minimize the discrepancy between the measured samples $\mathcal{X}_l$, and the samples generated from the solution after they pass through the forward imaging model $\mathcal{A}(\hat{\mathcal{X}}_h)$. Specifically, the compressed sensing problem for satellite lidars is formulated as

$$\min_{\hat{\mathcal{X}}_h} \left\| \mathcal{X}_l - \mathcal{A}(\hat{\mathcal{X}}_h) \right\|^2 + \lambda \mathbf{R}(\hat{\mathcal{X}}_h) \tag{7}$$

where λ represents a weight parameter and $\mathbf{R}$ denotes a regularization term that can be used to enforce sparsity in the solution within a certain basis or impose smoothness in the reconstruction. However, the forward imaging model for CS-Lidars exhibits stochastic behavior. Consequently, a Bayesian approach becomes more suitable for compressed sensing in this context. In Bayesian compressed sensing [28], we define a joint distribution $p(\mathcal{X}_l, \mathcal{X}_h)$ that encompasses all observed and unknown variables. Our goal is to recover the high-resolution tensor by maximizing this joint distribution

$$\max_{\hat{\mathcal{X}}_h} p(\mathcal{X}_l, \hat{\mathcal{X}}_h).$$

By further factorizing this joint distribution and optimizing over its logarithm, we arrive at the following expression:

$$\max_{\hat{\mathcal{X}}_h} \log p(\mathcal{X}_l | \hat{\mathcal{X}}_h) + \log p(\hat{\mathcal{X}}_h) \tag{8}$$

where the first term captures the likelihood of the satellite samples given the forward imaging model, i.e., the reconstruction term, and the second term represents our prior knowledge about the solution. We observe that maximizing (8) is equivalent to maximizing the posterior distribution $p(\hat{\mathcal{X}}_h | \mathcal{X}_l)$. This equivalence arises from the fact that the evidence term $p(\mathcal{X}_l)$ remains independent of the solution to the compressed sensing problem. Consequently, this approach aligns seamlessly with the principles of traditional Bayesian inference.

Given that our forward imaging model is described in terms of multinomial distributions over aggregated footprints, where each probability mass is given by (6), the likelihood term in (8) takes the form

$$p(\mathcal{X}_l | \mathcal{X}_h) = \prod_{i=1}^{n} \prod_{j=1}^{m} f\left(\mathcal{X}_l^{(i,j)}; r, (\mathcal{G} * \mathcal{X}_h)^{(i,j)} \right). \tag{9}$$

If we examine the maximum likelihood estimation of (9) for single a footprint, we notice that[1]

$$\begin{aligned} \max_{\mathbf{x}_a} \; f(\mathbf{x}_l; r, \mathbf{x}_a) &= \max_{\mathbf{x}_a} \; \frac{r!}{\prod_{i=1}^{c} x_l^i!} \prod_{i=1}^{c} \nu_i^{x_l^i}, \\ &= \max_{\mathbf{x}_a} \; \log r! - \sum_{i=1}^{c} \log x_l^i! + \sum_{i=1}^{c} x_l^i \log \nu_i \end{aligned}$$

where $\log r!$ and $\sum_{i=1}^{c} \log x_l^i!$ are independent of $\mathbf{x}_a$, and hence

$$\max_{\mathbf{x}_a} \; f(\mathbf{x}_l; r, \mathbf{x}_a) = \min_{\mathbf{x}_a} \; - \sum_{i=1}^{c} x_l^i \log \nu_i = \min_{\mathbf{x}_a} \; \underbrace{H(\mathbf{x}_l, \mathbf{x}_a)}_{\text{Cross-Entropy}} .$$

Since the maximum likelihood estimation for a single footprint can be formulated as a cross-entropy minimization, we can apply this reasoning to all footprints, rewriting (8) as

$$\min_{\hat{\mathcal{X}}_h} H\left(\mathcal{X}_l, \mathcal{G} * \hat{\mathcal{X}}_h\right) - \frac{1}{\lambda} \log p\left(\hat{\mathcal{X}}_h\right)$$

where $H(\cdot, \cdot)$ denotes the cross entropy. Moreover, using the definition

$$D_{\text{KL}}\left(\mathcal{X}_l \,\|\, \mathcal{G} * \hat{\mathcal{X}}_h\right) = H\left(\mathcal{X}_l, \mathcal{G} * \hat{\mathcal{X}}_h\right) - H(\mathcal{X}_l)$$

and noting that the optimization is independent of $\mathcal{X}_l$, we arrive at

$$\min_{\hat{\mathcal{X}}_h} \; \underbrace{D_{\text{KL}}\left(\mathcal{X}_l \,\|\, \mathcal{G} * \hat{\mathcal{X}}_h\right)}_{\text{Reconstruction}} - \frac{1}{\lambda} \underbrace{\log \; p\left(\hat{\mathcal{X}}_h\right)}_{\text{Regularization}} \tag{10}$$

where $D_{\text{KL}}(\cdot, \cdot)$ is the KL divergence. Similar to the effect of $\mathbf{R}$ in (7), the prior $p(\mathcal{X}_h)$ can be used to regularize the solution such that it is sparse in some basis, e.g., assuming a Laplace prior [28]. In the context of this work, we impose constraints upon the prior distribution using a generative diffusion model. To solve this optimization problem, we employ the DPS algorithm [29], which iteratively optimizes the likelihood term while constraining the solution to adhere to the distribution defined by a diffusion generative model, as described in Sections V-B and V-D. The outcome of this process yields super-resolved solutions that resemble authentic high-resolution tensors.

IV. Denoising Diffusion Probabilistic Models

DDPMs are a class of generative models that can produce high-quality images by simulating a diffusion process that gradually adds noise to the data until it becomes pure noise and then reverses the process by denoising the data until it recovers the original image.

A. Forward Diffusion Process

The forward diffusion process is a Markov chain representing a stochastic process that gradually corrupts the original data into random Gaussian noise (Fig. 7). Let $\mathbf{x}_0$ be the original data sample, such as an image or an HHDC, drawn from the data distribution $p_{\text{data}}(\mathbf{x}_0)$. The forward diffusion process transforms $\mathbf{x}_0$ into a sequence of noisy samples $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T$,

[1] Strictly speaking, the likelihood function for a multinomial distribution is valid only when $\boldsymbol{\nu}$ represents a proper probability distribution. Therefore, it is essential to normalize $\boldsymbol{\nu}$ by scaling it with the sum of its elements.

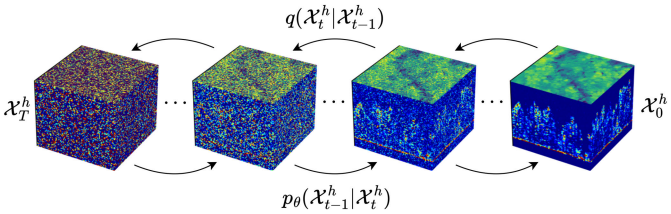


Fig. 7. Denoising diffusion methodology. Starting from a tensor of pure noise $\mathcal{X}_T^h$, the system learns to reconstruct the high-resolution HHDC $\mathcal{X}_0^h$ by following a series of learned denoising steps described by a neural network p_θ.

where T is the number of diffusion steps. At each step t, the process adds Gaussian noise to the previous sample $\mathbf{x}_{t-1}$ according to the following distribution:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right)$$

which can be rewritten using a simple reparametrization as

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\ \boldsymbol{\epsilon}_t$$

where $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is a standard normal random variable and β_t is a predefined variance schedule that controls the amount of noise added at each step. It is noteworthy to mention that the forward diffusion process allows us to sample $\mathbf{x}_t$ at any arbitrary timestep t in closed form starting from $\mathbf{x}_0$ as

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, \left(1-\bar{\alpha}_t\right)\mathbf{I}\right)$$

where $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$.

B. Reverse Diffusion Process

The reverse diffusion process starts from a pure noise distribution $p(\mathbf{x}_T)$, where T is the total number of diffusion steps, and gradually removes the noise to reconstruct the original signal $\mathbf{x}_0$ (Fig. 7). At each step t, the model samples from a conditional distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$, where $\boldsymbol{\mu}_\theta$ and $\boldsymbol{\Sigma}_\theta$ are neural networks with learned parameters. The reverse process can then be written as

$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t).$$

To sample from the reverse process, we first draw a pure noise sample $\mathbf{x}_T \sim p(\mathbf{x}_T)$ and then use the following recursive formula to obtain $\mathbf{x}_{t-1}$ from $\mathbf{x}_t$:

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)\boldsymbol{\epsilon}_t$$

where $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is an independent noise term. In this way, we can generate samples from $p(\mathbf{x}_{0:T})$ by applying the reverse process from T to 0.

C. Training Objective

The training objective is to minimize the negative log likelihood of the data given the noise or, equivalently, to minimize the KL divergence between the data distribution and the model distribution at each timestep. The training objective can then be written as

$$\mathbb{E}\left[-\log p_\theta(\mathbf{x}_0)\right] \leq \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

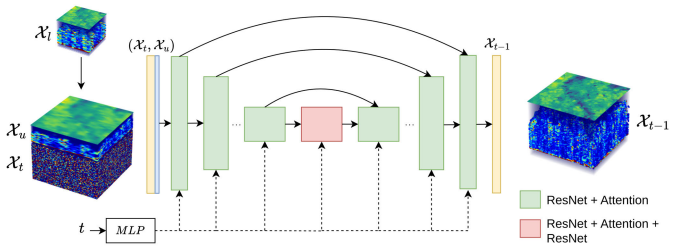


Fig. 8. U-Net-like architecture that allows to approximate the conditional probability distribution $p_\theta(\mathcal{X}_{t-1}^h|\mathcal{X}_t^h, \mathcal{X}^l)$. It is important to realize that this neural network receives three different inputs: the noisy tensor at time t, the variable t itself, and the low-resolution tensor upscaled and concatenated to the noisy input. The network implements ResNet-type blocks followed by attention blocks in both the encoder and the decoder and also has a middle block with a double ResNet to prepare the input for the decoder. In total, the network performs downsampling/upsampling in four stages with base feature maps of 128 and depth multipliers of (8,16,16,16), with eight ResNet blocks in each stage. By further concatenating additional signals (such as a monochromatic image), we can incorporate more sources of side information into the system.

where the expectation is bounded because of the usual variational lower bound. Using the chain rule and Bayes' rule, we can rewrite the training objective as

$$\mathcal{L}(\theta) = -\mathbb{E}_q\left[\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^{T}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right]$$

where $p_\theta(\mathbf{x}_T)$ is the prior distribution of the noise, which is usually assumed to be Gaussian. The second term in the objective is the sum of log-likelihood ratios between the reverse and forward transition distributions, which can be interpreted as a measure of how well the model can denoise the data at each step. The objective can be optimized using stochastic gradient descent, by sampling a mini-batch of data and applying the forward and reverse processes to obtain the intermediate states $\mathbf{x}_t$.

D. Neural Network Architecture

In DDPMs, the U-Net architecture is widely used due to its ability to predict the parameters of the reverse diffusion process while maintaining the spatial structure of the data (Fig. 8). U-Net is particularly effective for image generation tasks, as it allows the model to denoise corrupted data across different scales, making it ideal for progressively refining images in the reverse diffusion process. The U-Net architecture consists of K hierarchical downsampling and upsampling blocks, designed to capture both local and global features. Each downsampling block reduces the resolution of the input by a factor of 2, while the corresponding upsampling block restores it. Skip connections link each downsampling block with its corresponding upsampling block, ensuring that high-resolution details from earlier layers are preserved throughout the reverse diffusion process. This mechanism is crucial for accurate image generation, as it allows the model to retain fine-grained spatial information that would otherwise be lost during downsampling. Each block in the U-Net includes a ResNet block, responsible for refining the feature representations, followed by an attention mechanism. The attention layers enable the model to focus on specific regions of the image that are most relevant at each stage of denoising. This is especially important

in diffusion models, where the network needs to effectively target areas of the image that have been heavily corrupted by noise. A middle block, positioned at the bottleneck of the U-Net, combines ResNet and attention layers to process the lowest resolution features, further enhancing the network's ability to learn complex patterns. A key aspect of the U-Net's effectiveness in diffusion models is its ability to incorporate scalar conditioning, such as the diffusion timestep t. The timestep is embedded into intermediate layers of the U-Net, allowing the network to adjust its denoising predictions based on the current stage of the reverse process. This conditioning mechanism ensures that the model can learn a time-dependent mapping from noisy data to clean images, effectively guiding the reverse diffusion trajectory. By utilizing this structure, the U-Net efficiently predicts the parameters $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$ and $\boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)$ at each diffusion step, directly influencing the reverse process described in Section IV-B. Its architectural design aligns with the variational training objective, enabling the model to minimize the difference between the forward and reverse distributions across timesteps, ultimately generating high-quality samples from noisy data.

V. Guiding Diffusion Models

Diffusion models are a powerful tool for generating high-resolution HHDC, but they are self-supervised models focused solely on learning the distribution of the training data. As a result, they can only produce random samples that resemble the original dataset. To address super-resolution challenges in satellite lidars, we must develop methods that guide these models beyond random generation. In this work, we intend to guide diffusion models using three different strategies. First, we will guide the diffusion model by explicitly learning the conditional distribution between the low-resolution satellite measurements and the high-resolution output. This implies that the model goes from being completely unsupervised to having some supervision. Second, we will leverage the inverse problem presented in Section III so that the gradient of the likelihood term in cost function (8) gives us a direction in the reverse diffusion process. Finally, we will guide the diffusion model using side information that could potentially be installed in the satellite. In particular, we will use a high-resolution monochromatic image that has the spatial resolution of the high-resolution tensor.

A. Explicit Conditional Probability Estimation

In conditional generation scenarios, the data carry a complementary conditioning signal, denoted as $\mathbf{c}$. This signal could embody a class label for class-conditional generation, a low-resolution image for super-resolution tasks, or other relevant context-setting information. The aim is to construct a conditional model $p_\theta(\mathcal{X}_0|\mathbf{c})$, capable of generating data conditioned on the signal $\mathbf{c}$ [23], [30]. To achieve this, we introduce $\mathbf{c}$ as an input to the reverse diffusion process as follows:

$$p(\mathcal{X}_{0:T}|\mathbf{c}) = p(\mathcal{X}_T)\prod_{t=1}^{T} p_\theta(\mathcal{X}_{t-1}|\mathcal{X}_t, \mathbf{c}) \tag{11}$$

where

$$p_\theta(\mathcal{X}_{t-1}|\mathcal{X}_t, \mathbf{c}) = \mathcal{N}\big(\mathcal{X}_{t-1}; \boldsymbol{\mu}_\theta(\mathcal{X}_t, t, \mathbf{c}), \boldsymbol{\Sigma}_\theta(\mathcal{X}_t, t, \mathbf{c})\big). \tag{12}$$

In this framework, the data and conditioning signal $(\mathcal{X}_0, \mathbf{c})$ are sampled together from the data distribution, denoted as $p(\mathcal{X}_0, \mathbf{c})$. The forward process $q(\mathcal{X}_{1:T}|\mathcal{X}_0)$ remains unaltered. The key modification lies in incorporating $\mathbf{c}$ as an additional input to the neural network p_θ so that now $\boldsymbol{\mu}_\theta$ and $\boldsymbol{\Sigma}_\theta$ become $\boldsymbol{\mu}_\theta(\mathcal{X}_t, t, \mathbf{c})$ and $\boldsymbol{\Sigma}_\theta(\mathcal{X}_t, t, \mathbf{c})$, respectively. The specific architectural decisions for incorporating these inputs depend on the nature of the conditioning signal $\mathbf{c}$. For lower resolution HHDC conditioning $\mathcal{X}_l$, the model employs height-wise concatenation, meticulously combining the low-resolution tensor (which has undergone bilinear upsampling to attain the desired spatial resolution) with the reverse process input $\mathcal{X}_t$ (Fig. 8).

B. Inverse Problem Guidance

In traditional diffusion models, the reverse diffusion process aims to reconstruct data by following directions where the data distribution $p_t(\mathcal{X}_t)$ is most concentrated. This is achieved using the *score function* $\nabla_{\mathcal{X}_t} \log p_t(\mathcal{X}_t)$, which points in the direction of the steepest ascent in the log-probability density. By taking small steps along the score function, we are more likely to reach states with higher probability. When incorporating measurements $\mathbf{c}$—which, in the context of satellite lidars, correspond to the low-resolution tensor $\mathcal{X}_l$—it is natural to consider the conditional distribution $p_t(\mathcal{X}_t|\mathbf{c})$ to guide the diffusion process. Applying Bayes' theorem yields

$$p_t(\mathcal{X}_t|\mathbf{c}) = \frac{p_t(\mathbf{c}|\mathcal{X}_t)\, p_t(\mathcal{X}_t)}{p_t(\mathbf{c})}.$$

Taking the gradient of the logarithm with respect to $\mathcal{X}_t$, we obtain

$$\nabla_{\mathcal{X}_t} \log p_t(\mathcal{X}_t|\mathbf{c}) = \underbrace{\nabla_{\mathcal{X}_t} \log p_t(\mathcal{X}_t)}_{\text{prior}} + \underbrace{\nabla_{\mathcal{X}_t} \log p_t(\mathbf{c}|\mathcal{X}_t)}_{\text{likelihood}}.$$

This expression indicates that the reverse diffusion direction depends on both the prior term (the unconditional distribution) and a likelihood term involving $p_t(\mathbf{c}|\mathcal{X}_t)$. However, calculating the likelihood term is challenging because the dependence between the measurements and the signal exists at every intermediate time step t, and we typically only have knowledge of the forward measurement model at $t = 0$, namely, $p_0(\mathbf{c}|\mathcal{X}_0)$.

To address this issue, we modify the DPS algorithm proposed in [29] by incorporating our own likelihood term derived from our forward imaging model. Specifically, [29] approximates the likelihood term using

$$p_t(\mathbf{c}|\mathcal{X}_t) \approx p\big(\mathbf{c}|\hat{\mathcal{X}}_0\big)$$

where

$$\hat{\mathcal{X}}_0 := \mathbb{E}[\mathcal{X}_0|\mathcal{X}_t] = \mathbb{E}_{\mathcal{X}_0 \sim p(\mathcal{X}_0|\mathcal{X}_t)}[\mathcal{X}_0]$$

is the estimated clean signal at $t = 0$ obtained from $\mathcal{X}_t$. By leveraging our forward imaging model in this approximation, we define a likelihood term that ensures data fidelity through the KL divergence between the measurements $\mathbf{c}$ (the low-resolution tensor $\mathcal{X}_l$ in our context) and the high-resolution tensor $\hat{\mathcal{X}}_0$ after it has been processed by our model. The modified likelihood gradient becomes

$$\nabla_{\mathcal{X}_t} \log p_t(\mathbf{c}|\mathcal{X}_t) \approx \nabla_{\mathcal{X}_t} \log p\big(\mathbf{c}|\hat{\mathcal{X}}_0\big)$$

Algorithm 1 DPS [29] With KL Divergence

Require: N, $\mathcal{X}_l$, $\{\zeta_i\}_{i=1}^N$, $\{\tilde{\sigma}_i\}_{i=1}^N$
1: $\mathcal{X}_h^{(T)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $i = N-1$ **to** 0 **do**
3: $\quad \hat{\mathbf{s}} \leftarrow \mathbf{s}_\theta(\mathcal{X}_h^{(i)}, i)$
4: $\quad \hat{X}_h^{(0)} \leftarrow \frac{1}{\sqrt{\bar{\alpha}_i}}\left(\mathcal{X}_h^{(i)} + (1-\bar{\alpha}_i)\hat{\mathbf{s}}\right)$
5: $\quad \mathcal{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
6: $\quad \mathcal{X}'^{(i-1)}_h \leftarrow \frac{\sqrt{\alpha_i}(1-\bar{\alpha}_{i-1})}{1-\bar{\alpha}_i}\mathcal{X}_h^{(i)} + \frac{\sqrt{\bar{\alpha}_{i-1}}\beta_i}{1-\bar{\alpha}_i}\hat{X}_h^{(0)} + \tilde{\sigma}_i \mathcal{Z}$
7: $\quad \mathcal{X}_h^{(i-1)} \leftarrow \mathcal{X}'^{(i-1)}_h - \zeta_i \nabla_{\mathcal{X}_l}[D_{\mathrm{KL}}(\mathcal{X}_l \,||\, \mathcal{G} * \hat{X}_h^{(0)})]$
8: **end forreturn:** $\mathcal{X}_h^{(0)}$

which is tractable since it involves the known forward model at $t = 0$. By integrating this modified likelihood term into the reverse diffusion process, as detailed in Algorithm 1, we ensure that our diffusion model not only follows the data distribution but also maintains fidelity to the measurements. This is achieved by minimizing the KL divergence between the measurements and the output of our forward imaging model applied to $\hat{\mathcal{X}}_0$, thus guaranteeing that the reconstructed data align closely with the observed measurements.

C. Side-Information Guidance

Frequently, sensors deployed in orbit utilize multimodal technologies, wherein multiple signals from different sensors are captured concurrently, enabling data fusion. An exemplary case will be CASALS, which will simultaneously capture lidar and hyperspectral images to enhance the information obtained from the terrain. Consequently, it is logical to extend the concept of super-resolution to scenarios involving, at a minimum, grayscale images of the terrain under analysis. Accordingly, it is imperative to develop strategies that leverage this additional information within the framework of diffusion model methodologies. Fortunately, the way we can include additional information is entirely analogous to how we studied the explicit estimation of a conditional probability distribution in Section V-A. The only difference is that now, when performing the reverse process in (11), each individual step is additionally conditioned on any side information $\mathbf{s}$ that the satellite may acquire such that

$$p(\mathcal{X}_{0:T}|\mathbf{c}, \mathbf{s}) = p(\mathcal{X}_T)\prod_{t=1}^{T} p_\theta(\mathcal{X}_{t-1}|\mathcal{X}_t, \mathbf{c}, \mathbf{s}). \tag{13}$$

Similarly, this new signal can be incorporated into the U-Net through a channel-wise concatenation. It is important to highlight that this latest proposed methodology is an extension of the methodology proposed in Section V-A, and thus, it is also necessary to include the low-resolution tensor as input.

D. Combined Inverse Problem and Side-Information Guidance

Finally, it is important to add that it is possible to combine all the aforementioned methodologies into a framework that leverages both side information and the information from a forward imaging model. By utilizing side information, such as multimodal sensor data, and incorporating a forward model, we can more effectively guide the diffusion process toward the desired solution.

In the traditional inverse problem setup for diffusion models, the reverse process is guided by both the prior distribution $p_t(\mathcal{X}_t)$ and a likelihood term $p_t(\mathbf{c}|\mathcal{X}_t)$. As derived earlier, the likelihood term can be approximated using a forward model evaluated at the estimated clean signal $\hat{\mathcal{X}}_0$, leading to

$$p(\mathbf{c}|\mathcal{X}_t) \simeq p\left(\mathbf{c}|\hat{\mathcal{X}}_0\right) \tag{14}$$

where now $\hat{\mathcal{X}}_0 = \mathbb{E}_{\mathcal{X}_0 \sim p(\mathcal{X}_0|\mathcal{X}_t, \mathbf{s})}[\mathcal{X}_0]$ depends on the side information $\mathbf{s}$. This forward model-based guidance using side information allows the diffusion model to incorporate knowledge of the measurement process and achieve higher fidelity reconstructions while being aware of additional sources of data.

In practice, this additional conditioning is implemented by extending the architecture of the score model (e.g., U-Net) to include the side information $\mathbf{s}$ through a channel-wise concatenation. This approach allows the model to leverage both the forward imaging model and the additional side information, improving the quality and accuracy of the generated solutions. Furthermore, as in the inverse problem framework, the clean signal estimation $\hat{\mathcal{X}}_0$ can still be used to guide the likelihood term during the reverse process, ensuring that both prior knowledge and measurement consistency are preserved.

This combined framework offers a flexible and powerful method for solving inverse problems in scenarios where additional information is available. The inclusion of side information not only enhances the model's ability to reconstruct fine details but also improves robustness to noise and other uncertainties in the data. As such, this approach represents a natural extension of the methodologies discussed in Sections V-A–V-C, aligning with the overarching goal of developing efficient and reliable diffusion models for complex, real-world tasks.

VI. Dataset Description and Training Setup

Recent advances in generative models have enabled significant progress in the reconstruction of high-resolution data from sparse and compressed observations. To evaluate the efficacy of our proposed methodology, we construct a controlled experimental setup using emulated data, designed to mimic the characteristics of satellite lidar systems. Fig. 9 presents an overview of the proposed methodology, highlighting the components that leverage emulated HHDCs derived from the datasets we generated. This approach ensures a rigorous assessment of the reconstruction capabilities under realistic conditions.

A. Dataset Emulation

A dataset of emulated HHDCs was created using high-resolution point clouds from G-LiHT and NEON, following the methodology explained in Section II. The source point clouds were extracted from two distinct regions across USA that are more than 1500 km apart: the Everglades National Park in the state of Florida, centered at latitude 25.75°N and longitude 80.56°W, and the SERC in the state of Maryland, centered at latitude 38.88°N and longitude 76.56°W. The input low-resolution HHDC footprints were

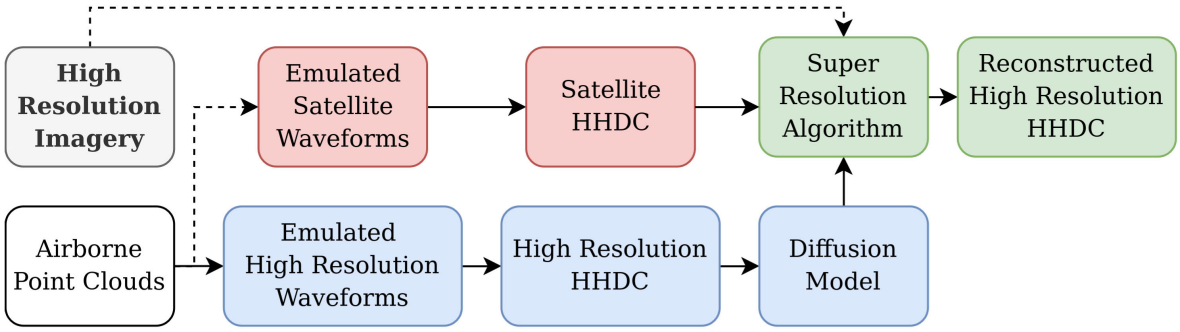


Fig. 9. Overview of the main methodology. We start with high-resolution airborne point cloud measurements to emulate two distinct waveforms and high-resolution HHDCs tailored to the desired resolution. As input, we simulate satellite waveforms and HHDCs matching the future CASALS lidar specifications—10-m-diameter footprints distributed on a grid with 6-m across-track and 3-m along-track spacing, and a vertical resolution of 0.5 m. Our output enhances these data to higher resolutions, such as 1-m footprints on a 1 × 1 m grid, using a super-resolution algorithm augmented with high-resolution grayscale images fed into neural networks. In the figure, blue blocks represent processes implemented entirely offline, red blocks will be replaced by real measurements from future satellite missions, and green blocks denote processing required either in-orbit or on the ground after data downlink.

designed with a uniform diameter of 10 m, spaced 3 m apart along the swath and 6 m across the swath, with a vertical resolution of 0.5 m. These HHDCs were standardized to a fixed size of 16 × 32 × 64 (footprints across the swath × footprints along the swath × height), corresponding to a forest tile covering 96 × 96 m with a height of 32 m. These parameters were selected to align with the resolution of a future CASALS mission. The high-resolution output HHDCs were designed to increase the resolution while keeping the covered area unchanged. For example, for super-resolution down to 2 m, the output HHDC footprints have a radius of 2 m, a 2-m separation both along and across the swath, and a vertical resolution of 0.25 m, resulting in a tensor size of 48 × 48 × 128 while maintaining the same 96 × 96 m coverage. The algorithms were tested for super-resolution at scales of 3, 2, and 1 m, using vertical resolutions of 0.25 m for the Florida dataset and 0.5 m for the SERC dataset.

For each instance in this dataset, three distinct components were created. First, a high-resolution HHDC was generated, containing hundreds to thousands of photons per footprint. Second, a low-resolution HHDC was produced, reflecting the photon loss characteristic of the CS-Lidar imaging model, with approximately 20 photons per footprint. Finally, a monochromatic image was extracted—either from the hyperspectral image (for G-LiHT) or from the RGB image (for NEON)—and adjusted to match the spatial resolution of the HHDC. In this setup, each spectral pixel corresponds to the same area as the high-resolution footprint.

B. Training Setup

To ensure the reproducibility of our methodology, we provide a detailed description of the training setup for the diffusion model employed in this study, which follows the DDPM framework. The training process utilized approximately 12 000 data cubes for each region of interest. To enhance the robustness of the model and mitigate overfitting, data augmentation was incorporated into the preprocessing pipeline. Specifically, random vertical and horizontal flips were applied to the training data, introducing variability while preserving the underlying statistical characteristics of the dataset. The model architecture used in this study is the U-Net described in Section IV-D. This architecture is a common choice for diffusion models due to its ability to effectively capture complex dependencies in high-dimensional data. In our setup, the U-Net was employed to learn the score function of the data distribution, as is traditional in diffusion frameworks. The diffusion process was modeled with a total of 1000 steps, providing a high level of granularity in the learned representation. The training was conducted on an NVIDIA A100 GPU with 80 GB of memory, leveraging its computational capabilities to expedite the training process. The model was trained using a batch size of 64 and iterated over 500 000 mini-batches to ensure adequate convergence. The learning rate for the DDPM was set to 8×10^{-5}, a value chosen based on standard practices for training diffusion models. The Adam optimizer, with $\beta_1 = 0.9$ and $\beta_2 = 0.99$, was used to minimize the loss function, ensuring stable and efficient convergence. To further stabilize the training process, we applied an exponential moving average (EMA) to the model's parameters with a decay factor of 0.995. The EMA updates were performed every ten training steps, enabling smoother convergence and improved generalization of the model. To promote reproducibility, all hyperparameters were initialized according to the configurations recommended in the DDPM framework, with minimal adjustments tailored to this specific application. Furthermore, the codebase, including all training scripts and data preprocessing pipelines, has been made available in a publicly accessible repository to enable the community to replicate the results presented in this work. This detailed description of the training process highlights the steps taken to ensure transparency and reliability, thereby providing a foundation for others to replicate and extend the findings of this study.

VII. Results and Discussion

A. Florida Everglades National Park

The initial study area is a densely forested region located in the southern part of Florida, encompassing approximately 400 by 600 m. Figs. 10 and 11 illustrate the comparative analyses of the proposed algorithms at 3- and 1-m spatial resolutions. Notably, reconstructions at 1 m using the model incorporating inverse problem guidance and auxiliary side information require an average computation time of 50 min/reconstruction. It is important to note that a 3-m

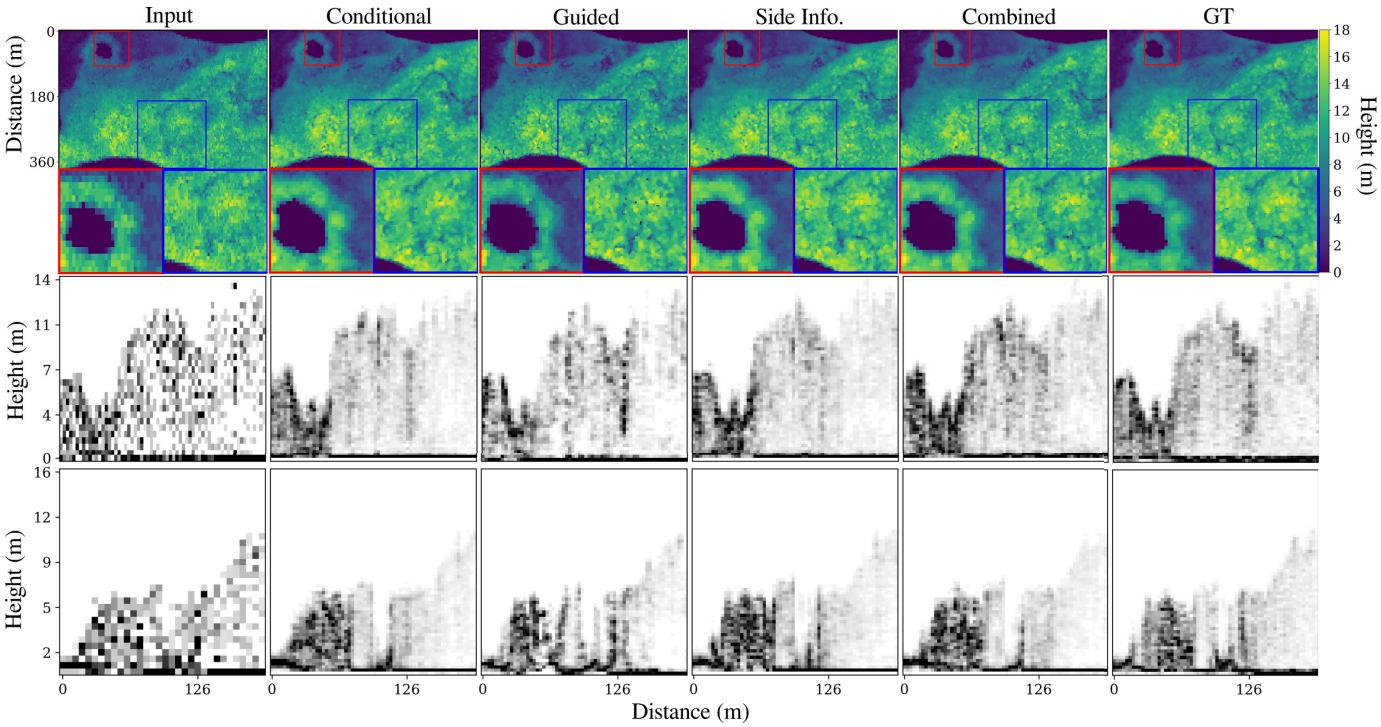


Fig. 10. Comparison of the super-resolution algorithm based on diffusion models for a forested area in Florida, in scenarios with and without side information, as well as those incorporating guidance using an inverse problem. The input is an HHDC with footprints of 5 m radius, spaced 6 × 3 m apart, and a vertical resolution of 0.5 m. The output consists of HHDC with footprints of 3 m radius, spaced 3 × 3 m apart, and a vertical resolution of 0.25 m. The six columns in the figure show: the low-resolution input, the output without side information, the output with inverse problem guidance, the output with side information, the output with side-information and inverse problem guidance, and the GT. The three rows display the CHM (with two zoomed-in views), a profile along the track, and a profile across the track. The grayscale values of the profiles represent the number of photons in each bin, which have been normalized for comparison purposes.

super-resolution has already been explored in [30] and [31]. Therefore, we conduct more comprehensive studies across a wider range of resolutions. In addition, Table I provides a qualitative analysis of the results, and Fig. 12 shows the absolute errors for all the proposed methodologies on the CHM. From both figures and the results table, it is evident that the algorithms based on side information deliver the best performance. Only in a few scenarios, can the inverse problem-guided system surpass the performance of the side information-based approach by itself.This outcome is expected, as the side-information system has access to additional data that the other algorithms have never seen. What is particularly important to highlight is that when guiding diffusion models with an inverse problem, it is possible to achieve competitive results with significantly smaller neural networks. This approach can match or closely approach the performance of side-information models while using considerably fewer computational resources. This finding suggests that the proposed method can be effectively applied in resource-constrained environments where the computational cost is a key concern. Moreover, it is crucial to mention that when training the diffusion model to be guided by an inverse problem, no satellite samples were utilized at any stage. This implies that the algorithm could be easily adapted to perform tasks such as inpainting or denoising, making it versatile for a range of image restoration applications.

It is also important to address the challenges observed with the DTM. The performance of all algorithms was relatively poor in this region, which can be attributed to the flat terrain. Diffusion models inherently struggle with replicating flat surfaces, as there is less structural information for the model to leverage. This limitation highlights the need for future research into more sophisticated techniques that can handle such challenging terrains more effectively. One potential avenue for improvement lies in incorporating additional convolutional layers in combination with the diffusion model. These layers could act as a smoothing mechanism to enhance the reconstruction of flat regions, such as DTMs, by enforcing spatial consistency. In addition, future investigations could explore how including more areas with flat terrain in the training datasets might improve the models' ability to reconstruct such features, providing a more balanced learning experience. Such enhancements, coupled with hybrid architectures and multiscale or hierarchical diffusion models, could address these issues more effectively. Furthermore, these results have broader implications for future research and practical applications. It is crucial to acknowledge that the performance of side information-based algorithms depends heavily on the availability of external data, which may not always be feasible in certain environments. Thus, future work should focus on improving the robustness of diffusion models in data-scarce situations, ensuring that they can deliver competitive performance even when side information is unavailable.

B. Smithsonian Environmental Research Center

The second study area is the SERC forest, located in Maryland, covering an area of approximately 580 × 580 m. Figs. 13 and 14 present a comparative evaluation of

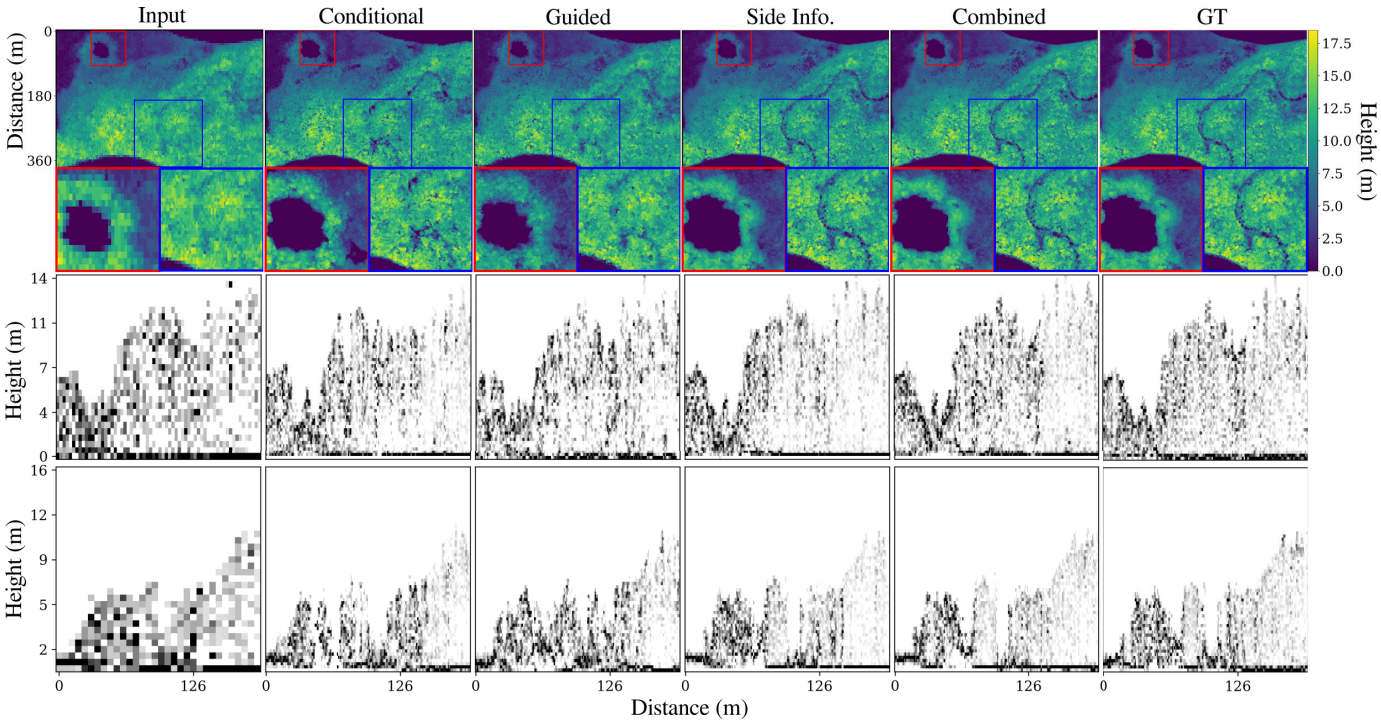


Fig. 11. Comparison of the super-resolution algorithm based on diffusion models for a forested area in Florida, in scenarios with and without side information, as well as those incorporating guidance using an inverse problem. The input is an HHDC with footprints of 5 m radius, spaced 6 × 3 m apart, and a vertical resolution of 0.5 m. The output consists of HHDC with footprints of 1 m radius, spaced 1 × 1 m apart, and a vertical resolution of 0.25 m. The six columns in the figure show the low-resolution input, the output without side information, the output with inverse problem guidance, the output with side information, the output with side-information and inverse problem guidance, and the GT. The three rows display the CHM (with two zoomed-in views), a profile along the track, and a profile across the track. The grayscale values of the profiles represent the number of photons in each bin, which have been normalized for comparison purposes.

TABLE I
COMPARISON OF THE PROPOSED APPROACHES ACROSS THREE REGIONS AND RESOLUTIONS

Resolution		3 meters				2 meters				1 meter			
Region	Metric	Conditional	Guided	Side	Combined	Conditional	Guided	Side	Combined	Conditional	Guided	Side	Combined
Florida	$PSNR_{chm}$	23.15	22.73	26.86	**27.23**	20.73	21.24	**27.15**	27.18	18.17	19.68	30	**30.26**
	$PSNR_{dtm}$	27.39	4.21	27.56	**27.29**	26.19	6.3	24.69	**30.1**	24.16	8.7	29.58	**29.83**
	$SSIM_{chm}$	0.63	0.69	**0.85**	**0.86**	0.46	0.55	**0.87**	**0.87**	0.26	0.32	**0.94**	**0.94**
	$SSIM_{dtm}$	0.9	0.31	**0.92**	0.91	0.9	0.31	0.92	**0.93**	0.85	0.28	**0.9**	**0.9**
SERC	$PSNR_{chm}$	25.55	25.4	28.95	**28.67**	23.28	24.25	**29.14**	28.92	18.25	20.35	23.89	**28.7**
	$PSNR_{dtm}$	27.72	31.11	**29.63**	29.08	25.56	30.84	29.81	**30.09**	24.06	26.18	25.26	**31.57**
	$SSIM_{chm}$	0.8	0.81	**0.9**	**0.89**	0.69	0.74	**0.91**	**0.9**	0.38	0.53	0.79	**0.89**
	$SSIM_{dtm}$	0.69	**0.75**	0.72	0.72	0.7	**0.79**	0.76	0.77	0.64	0.77	0.68	**0.82**
California	$PSNR_{chm}$	15.35	16.99	18.26	**19.03**	11.99	15.75	19.53	**20.19**	13.99	14.58	22.5	**23.08**
	$PSNR_{dtm}$	34.24	**35.88**	35.07	35.27	16.13	**35.37**	24.49	32.77	25.52	32.32	28.77	**35.31**
	$SSIM_{chm}$	0.65	0.71	0.79	**0.81**	0.33	0.61	0.85	**0.86**	0.35	0.41	0.89	**0.9**
	$SSIM_{dtm}$	0.86	**0.87**	0.85	**0.87**	0.75	**0.88**	0.84	0.87	0.85	0.87	0.87	**0.88**

the proposed algorithms at 3- and 1-m spatial resolutions within this region. Notably, reconstructions at 1-m resolution using the model with inverse problem guidance and auxiliary side information have an average computation time of 85 min/reconstruction. As seen in the Florida region, Table I provides a qualitative analysis of these results, and Fig. 15 shows the absolute errors for all the proposed methodologies on the CHM and the DTM. Notably, in contrast to the Florida findings, the inverse problem-guided diffusion model consistently outperforms the stand-along side-information algorithm in reconstructing DTMs in the SERC forest. This represents a significant shift, where the inverse problem-guided approach has now proven itself to be the most effective across the board for DTM reconstructions. One of the most striking findings from this region is that all algorithms, including those based on side information, are now able to reconstruct the DTM with relatively high accuracy. However, the inverse problem-guided model stands out as the top performer, delivering the most accurate and reliable DTM reconstructions across all tested resolutions. This suggests that in forested environments such as SERC, the inverse problem-guided model can take full advantage of the available structural information, even in

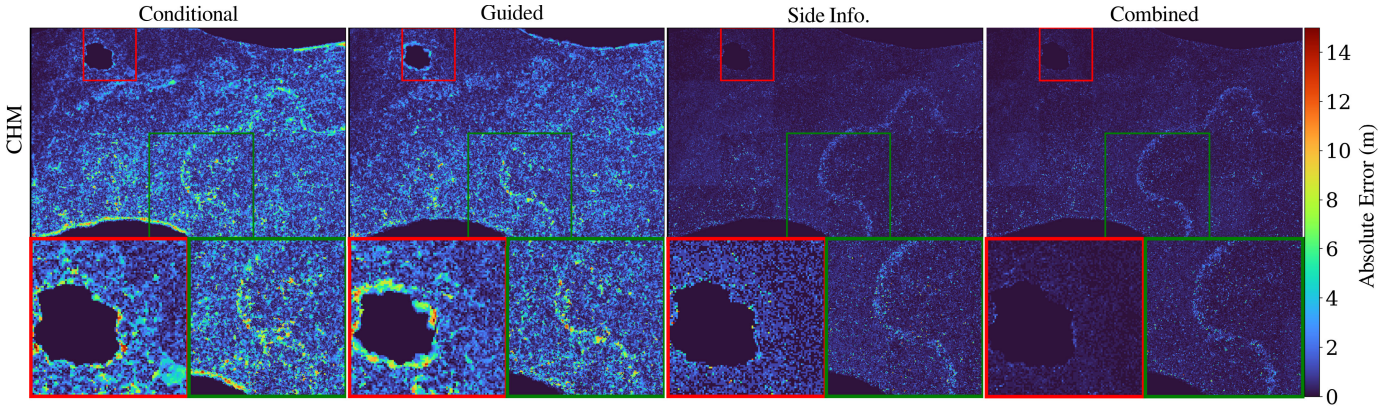


Fig. 12. Comparison of CHM absolute errors for a forested area in Florida, in scenarios with and without side information, as well as those incorporating guidance using an inverse problem. Darker shades of blue indicate a lower difference.

scenarios where side information is available to competing algorithms. The effectiveness of the inverse problem-guided model in this region is particularly interesting given the forest's dense canopy and varying terrain, which typically pose challenges for terrain reconstruction models. The inverse problem approach appears to handle these complexities more adeptly, potentially due to its ability to model underlying structures without relying on external data. The diffusion process in this model seems to excel at capturing the subtle topographical variations within the forest, whereas other models either rely heavily on side information or fall short in certain challenging areas. In terms of broader implications, the consistently superior performance of the inverse problem-guided model for DTM reconstructions suggests a potential shift in favor of simpler, more resource-efficient approaches in environments like SERC. The fact that the algorithm achieves this level of success without the need for extensive external data (as required by side information-based models) highlights its adaptability. This makes it particularly well suited for forested regions, where acquiring side information can be logistically difficult or resource-intensive.

To provide additional context, we also compared our proposed inverse problem-guided diffusion model with side information against two common super-resolution algorithms: CAE and interpolation, focusing on the challenging 1-m resolution reconstructions. Our results are summarized in Table II. While our approach consistently outperforms both in terms of accuracy, particularly in the reconstruction of CHMs under complex forest canopy conditions, it is important to acknowledge the significant differences in computational efficiency among the methods. Interpolation, as expected, is computationally trivial and can be performed almost instantaneously. Similarly, the CAE demonstrates computational efficiency, requiring less than 1 min/reconstruction. In contrast, our proposed approach, while achieving superior reconstruction accuracy, demands a substantially higher computational cost, with an average runtime of approximately 85 min/reconstruction at the 1-m resolution. This tradeoff between accuracy and computational cost underscores the distinct use cases and scenarios for which each algorithm may be best suited. Interestingly, the CAE exhibits strong performance in reconstructing DTMs, particularly in terms of maintaining consistency in less challenging regions. Its efficiency combined with its reasonable accuracy suggests potential for integration with more advanced methods. A promising avenue for future research could involve exploring hybrid approaches that leverage the computational efficiency of the CAE while incorporating the structural modeling strengths of the inverse problem-guided diffusion model. Such a combination might yield an optimized solution that balances accuracy and computational cost, particularly in environments like SERC, where terrain complexity and canopy density pose significant challenges.

C. California Lower Teakettle

Finally, to evaluate the zero-shot capabilities of the models, tests were conducted on an area that had never been seen during training: California. In this case, the newly introduced region was tested using the model trained on SERC, located over 2000 miles away. Table I presents the results of this experiment across multiple resolutions. As expected, performance degrades when attempting to super-resolve cubes that fall outside the original distribution. However, it is important to highlight that for the 1-m resolution, the results are quite remarkable for the combined algorithm using side information and the inverse problem framework. First, the PSNR for both the CHM and DTM exceeded 30 dB, which is competitive with various lossy compression algorithms (although still far from the performance achieved on the SERC distribution). Second, the SSIM for the CHM is equivalent to that of the original distribution, and the SSIM for the DTM is even higher. Upon examining the results for other resolutions and algorithms, this trend remains consistent across all metrics. It is evident that the DTM is more recoverable, as it contains fewer high-frequency components, making it more amenable to denoising algorithms such as diffusion models. Furthermore, guiding the models with the inverse problem once again proves to significantly enhance overall results. These zero-shot outcomes are promising and pave the way for a unified model that could potentially operate globally.

D. Opportunities for Improvement

While the methodologies presented in this study demonstrate substantial advancements in super-resolved satellite lidar

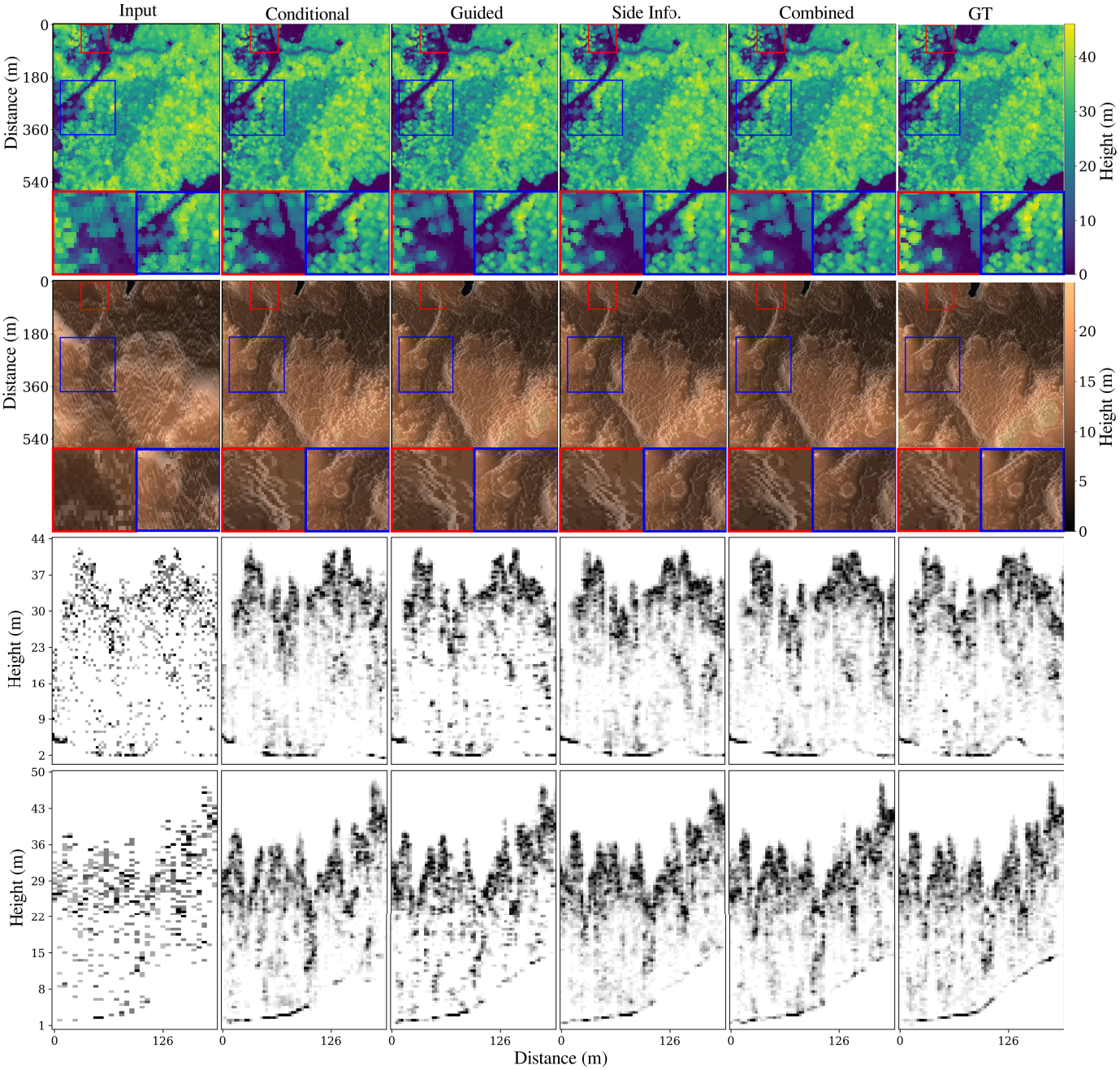


Fig. 13. Comparison of the super-resolution algorithm based on diffusion models for a forested area at the SERC, in scenarios with and without side information, as well as those incorporating guidance using an inverse problem. The input is an HHDC with footprints of 5 m radius, spaced 6 × 3 m apart, and a vertical resolution of 0.5 m. The output consists of HHDC with footprints of 3 m radius, spaced 3 × 3 m apart, and a vertical resolution of 0.5 m. The six columns in the figure show the low-resolution input, the output without side information, the output with inverse problem guidance, the output with side information, the output with side-information and inverse problem guidance, and the GT. The four rows display the CHM (with two zoomed-in views), the DTM (with two zoomed-in views), a profile along the track, and a profile across the track. The grayscale values of the profiles represent the number of photons in each bin, which have been normalized for comparison purposes.

imaging, they also reveal opportunities for further exploration and enhancement. Addressing these avenues can not only refine the proposed approaches but also extend their applicability to a broader range of scenarios and environments. A key area for improvement lies in the development of unsupervised learning techniques and domain adaptation methodologies. The current reliance on auxiliary high-resolution data, while effective, may not always be feasible in diverse and data-scarce environments. By enabling models to learn directly from unlabelled or sparsely labeled datasets, it becomes possible to achieve comparable performance with reduced dependence on high-resolution side information [32], [33], [34]. Domain adaptation can further enhance this process, ensuring that models trained on one region or sensor configuration generalize effectively to others, thus broadening their operational utility. Reconstructing flat terrains, such as DTMs, remains a challenging task due to the lack of distinctive structural features in these regions. Incorporating convolutional smoothing layers or hybrid architectures that blend traditional image processing techniques with advanced generative models may

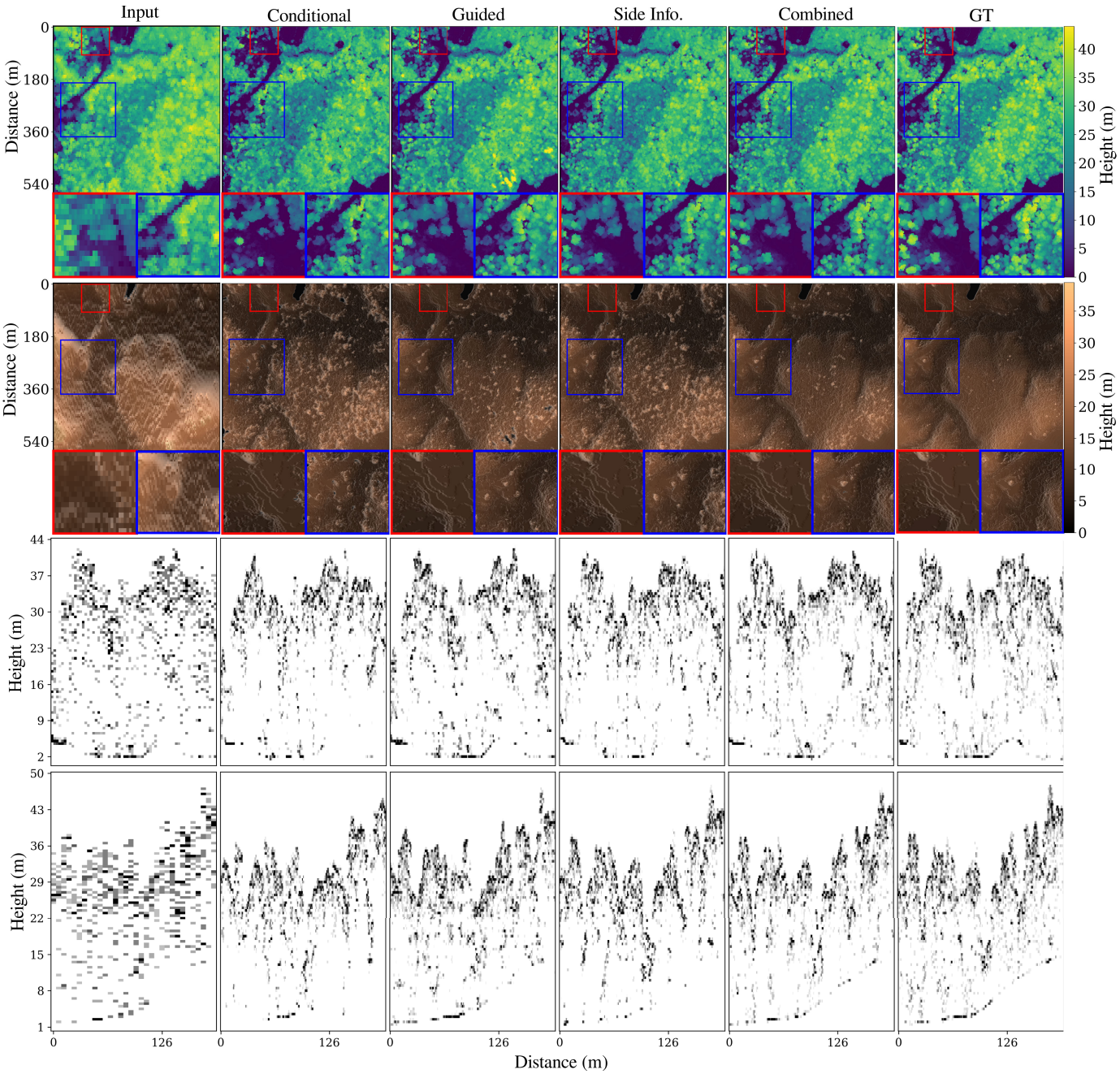


Fig. 14. Comparison of the super-resolution algorithm based on diffusion models for a forested area at the SERC, in scenarios with and without side information, as well as those incorporating guidance using an inverse problem. The input is an HHDC with footprints of 5 m radius, spaced 6 × 3 m apart, and a vertical resolution of 0.5 m. The output consists of HHDC with footprints of 1 m radius, spaced 1 × 1 m apart, and a vertical resolution of 0.5 m. The six columns in the figure show the low-resolution input, the output without side information, the output with inverse problem guidance, the output with side information, the output with side-information and inverse problem guidance, and the GT. The four rows display the CHM (with two zoomed-in views), the DTM (with two zoomed-in views), a profile along the track, and a profile across the track. The grayscale values of the profiles represent the number of photons in each bin, which have been normalized for comparison purposes.

offer a solution [9], [10]. These enhancements could enforce spatial consistency in reconstructed outputs, reducing artifacts and improving fidelity, particularly in regions where subtle variations in topography are critical. Another promising direction is the integration of transfer learning and synthetic data augmentation strategies. Transfer learning allows pretrained models to be fine-tuned on specific datasets, significantly reducing training times and resource requirements while improving performance on unseen data [35], [36]. Similarly, synthetic data augmentation, which generates diverse and realistic data variations, can improve robustness and model adaptability to new environments, ultimately leading to more reliable reconstructions in diverse geographic and ecological settings [37], [38]. Efficiency remains a cornerstone of operational satellite missions, and future efforts could focus on reducing computational costs. Latent diffusion models and other lightweight generative approaches hold significant promise in this regard [39]. By optimizing network archi-

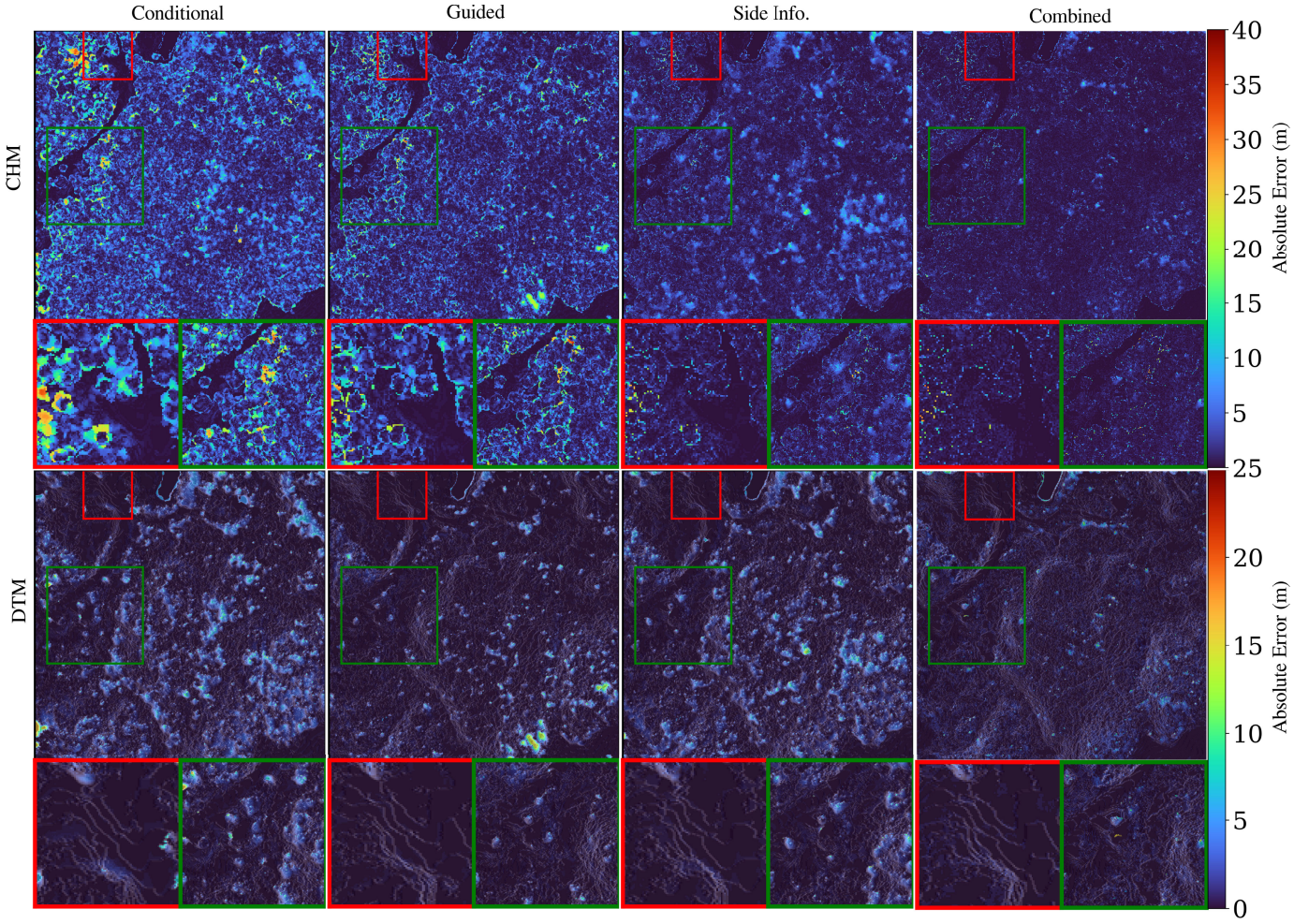


Fig. 15. Comparison of CHM and DTM absolute errors for a forested area at the SERC, in scenarios with and without side information, as well as those incorporating guidance using an inverse problem. Darker shades of blue indicate a lower difference.

TABLE II

COMPARISON OF THE PROPOSED APPROACHES WITH TWO BENCHMARK ALGORITHMS FOR 1-M SUPER-RESOLUTION ON SERC

1-meter Resolution	PSNR_{chm}	PSNR_{dtm}	SSIM_{chm}	SSIM_{dtm}
Interpolation	10.82	11.73	0.26	0.47
Autoencoder	16.03	30.05	0.36	0.82
Guided	20.35	26.18	0.53	0.78
Combined	**28.7**	**31.57**	**0.89**	**0.82**

tectures and leveraging model compression techniques, the computational burden can be mitigated without sacrificing performance. Such advancements would pave the way for real-time or near-real-time processing, making these models more suitable for in-orbit applications. Expanding the application domains of this framework represents another crucial frontier. While the current study focuses on forested regions, extending these methodologies to urban environments, atmospheric phenomena, or cryospheric studies would underscore their versatility and potential for addressing a wider array of scientific challenges. Such applications could inform urban planning, monitor greenhouse gas emissions, or track glacial dynamics, further demonstrating the transformative impact of super-resolved lidar imaging.

VIII. CONCLUSION AND FUTURE WORK

This article has presented a comprehensive framework for advancing super-resolved lidar imaging through innovative approaches that leverage diffusion models. By exploring three distinct methodologies, we have demonstrated a significant enhancement in the resolution of HHDCs of satellite lidar systems. The first approach, which learns the conditional probability distribution between low-resolution and high-resolution HHDCs, establishes a robust supervised training mechanism for the diffusion model. This method ensures that all features of the lidar shots are effectively utilized during the reverse denoising process. The second approach introduces a forward imaging model that provides a mathematical bridge between low-resolution and high-resolution data, guiding the diffusion model with closed-form expressions and gradients. This method mirrors techniques from compressed sensing to refine the resolution process. Finally, by incorporating side information with native resolution, we offer a valuable reference for the diffusion model, enhancing its capability to produce high-resolution outputs. Notably, the combination of inverse problem formulations guided by side information performs the best overall, offering a powerful synergy that maximizes the resolution and accuracy of the super-resolved outputs.

While our methods demonstrate exceptional performance, we acknowledge that relying on side information can be a limitation in scenarios where such data is not readily available, especially in remote or data-poor environments. This is one of the reasons why we chose to use grayscale images as side information, believing them to be among the most widely accessible forms of auxiliary data. Importantly, our results indicate that in many of the proposed scenarios where the super-resolution is not extreme, the inverse problem-guided method that does not rely on side information remains highly competitive. It exhibits only a few decibels decrease in performance compared to methods using side information and even surpasses them in certain metrics. Furthermore, future satellite missions such as CASALS are expected to integrate several concurrent imaging modalities alongside lidar, including spectral imaging, which could provide robust side information in the near future. This demonstrates that our approach maintains practical applicability in real-world situations while also being adaptable to advancements in satellite imaging technologies.

Our main contributions lie in the formulation of a novel forward imaging model tailored for satellite lidar systems, the development of a Bayesian compressed sensing framework to guide diffusion models, and the detailed methodology that facilitates significant improvements in HHDC resolution. Our results reveal that by leveraging inverse problem formulations to guide diffusion models, we achieve competitive results with considerably smaller network architectures. This efficiency not only reduces computational demands but also enhances the practicality of deploying these models in resource-constrained environments. In addition, we have explored various modifications to the guidance systems of diffusion models, highlighting their impact on super-resolution performance. The application of these methods to HHDCs of forested regions in USA underscores their practical utility and effectiveness. The implications of these advancements are far-reaching, providing a foundation for enhanced environmental monitoring, improved resource management, and more precise scientific research. The ability to produce higher resolution lidar images could significantly benefit fields such as forestry, agriculture, and urban planning by offering more detailed and accurate data.

Looking ahead, future work will focus on several promising directions. Expanding our methods to include latent diffusion models could offer even greater flexibility and performance in super-resolution tasks. Moreover, applying these techniques beyond forested areas to urban environments, and extending them to reconstruct clouds and aerosols, could greatly enhance our ability to monitor and understand complex and dynamic atmospheric and urban phenomena. Another critical area for future research is the development of better performance metrics that can be translated to out-of-distribution performance. Current metrics may not adequately capture the model's behavior when applied to data outside the training distribution, particularly in terms of hallucinations or artifacts introduced by the model. Quantifying the amount of hallucinations or spurious details generated by the model is essential for ensuring the reliability and robustness of super-resolved outputs in diverse environments. By further addressing the limitations identified during this study and exploring these new avenues, future research can refine these methods and open new possibilities for their application. This includes developing techniques that achieve high-resolution outputs without relying on side information, thereby increasing the applicability of our methods in data-poor environments. Our work provides a solid foundation upon which subsequent research can build, potentially leading to breakthroughs that will push the boundaries of high-resolution environmental data acquisition and analysis. Overall, the proposed approaches offer new avenues for advancing lidar imaging technology, with the potential to significantly enhance the clarity and precision of satellite-based observations. We anticipate that ongoing research and development will continue to advance these methods, contributing to a more detailed and actionable understanding of our environment.

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